Jianlei Huang

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INFORMATION

Website: https://jimmy000207.github.io/

EDUCATION

Georgetown University Washington, DC, USA
Ph.D. in Applied Mathematics *Aug. 2025-Present*

Georgetown University Washington, DC, USA
Master of Science in Mathematics and Statistics *Aug. 2023-May 2025*
Relevant courses: *Real Analysis (Ph.D. Level); Complex Variables (Ph.D. Level); Harmonic Analysis (Ph.D. Level); Partial Differential Equations (Ph.D. Level); Geometric Analysis (Ph.D. Level); Numerical Analysis (Ph.D. Level); Intro Partial Differential Equations; Probability; Mathematical Statistics; Deterministic Math Models*

Fudan University Shanghai, China
Bachelor of Engineering in Microelectronic Science and Engineering *Sept. 2018-June 2023*
Relevant courses: *Mathematical Analysis (3 semesters); Linear Algebra; Advanced Algebra (2 semesters); Functions of Complex Variable; Probability and Mathematical Statistics (Honorable Level)*

RESEARCH & PROJECT EXPERIENCE

Well-posedness for the Cubic Nonlinear Schrödinger Equation With Nonlinearity Supported on a Hypersurface
Supervised by Professor Benjamin Harrop-Griffiths from Georgetown University *Sept. 2025-Present*
- Derived a suitable Strichartz estimate for our equation.
- Proved a local well-posedness result in anisotropic Sobolev spaces on $\mathbb{R}^n$ using the contraction mapping.
- Proved the global well-posedness in the energy space for the defocusing equation in dimension 2 using dispersive estimates and energy conservation.
- Paper in preparation.

Gaussian Priors for Boundary Value Problems of Linear Partial Differential Equations
Collaborated Work with Professor Bogdan Raiţă from Georgetown University *Dec. 2023-Feb. 2025*
- Developed a new basis for the solution space derived from the Ehrenpreis-Palamodov Theorem.
- Provided formal proofs of correctness.
- Provided empirical results demonstrating significant accuracy and computational resource improvements over state-of-the-art approaches.
- Paper is currently under review at ICLR 2026.

On Non-symmetric Heisenberg Group in High Dimension
Tutorial research with Professor Der-Chen Chang from Georgetown University *Sept. 2025-Present*
- Derived the Sub-Laplacian and solved the Hamilton-Jacobi equations.
- Derived and counted geodesics.

PUBLICATIONS & PREPRINTS

- (with X. Chen) Well-posedness for the cubic nonlinear schrödinger equation with nonlinearity supported on a hypersurface. *In preparation, 2025.*
- (with M. Härkönen, M. Lange-Hegermann, and B. Raiţă) Gaussian process priors for boundary value problems of linear partial differential equations. *arxiv:2411.16663. Submitted, 2025.*

TEACHING

TA, Georgetown University
- MATH 3320: Functions of Complex Variables. Spring 2026.
- MATH 2410: Ordinary Differential Equations. Fall 2025.
- MATH 8130: Complex Variables. Spring 2025.
- MATH 8100: Real Analysis. Fall 2024.
- MATH 2430: Convex Geometry, Functions & Applications. Spring 2024.

PRESENTATION

APMA Doctoral Seminar — Georgetown University, Washington, D.C.

Topic: GP for Boundary Value Problems of Linear Partial Differential Equations — Oct. 11, 2024

CONFERENCE ATTENDED

Potomac Regional Analysis Seminar — Weekly, Fall 2025

Prairie Analysis Seminar 2025 — Nov. 7-8, 2025

Prairie Analysis Seminar 2024 — Oct. 25-26, 2024

Mid-Atlantic Regional Math Alliance Conference — Apr. 20, 2024

AMS 2024 Spring Eastern Sectional Meeting — Apr. 6-7, 2024

AWARDS & TECHNICAL PROFICIENCIES

- 2021 Interdisciplinary Contest in Modeling: *Finalist (top 1%)* — *Feb. 2021*
- 2020 China University Mathematics Contest in Modeling: *Second Prize* — *Sept. 2020*
- 2020 Interdisciplinary Contest in Modeling: *Meritorious Winner (top 5%)* — *Feb. 2020*
- Fudan University COVID-19 Mathematical Modeling Contest: *Second Prize* — *Jan. 2020*
- Technical skills: Latex, Python, R, C++, C, MATLAB
- Languages: Mandarin (native), English (proficient)